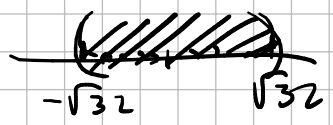
$B = \{ x \in \mathbb{R} / \quad x^2 - 5 < 27 \}$

$x^2 < 32$



$B = \{-5, -4, -3, \ldots, 0, \ldots, 3, 4, 5\}$

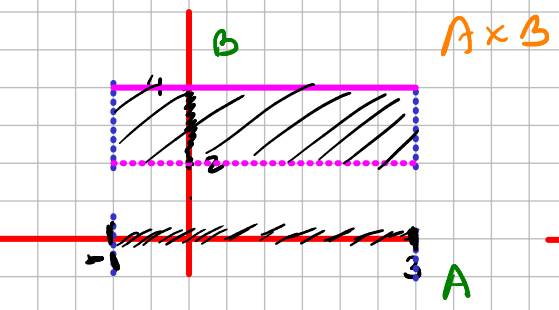
$n(A) = 8$

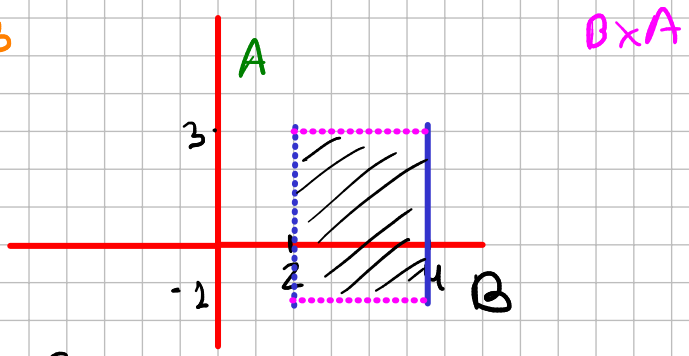
$n(B) = 11$

$n(A \times B) = n(A) \cdot n(B) = 8 \cdot 11 = 88$

④ $A = \langle -1; 3 \rangle$ $\quad A \times B \quad , \quad B \times A$

$B = \langle 2; 4]$ $\quad A = \{x \in \mathbb{R} / -1 < x < 3\}$





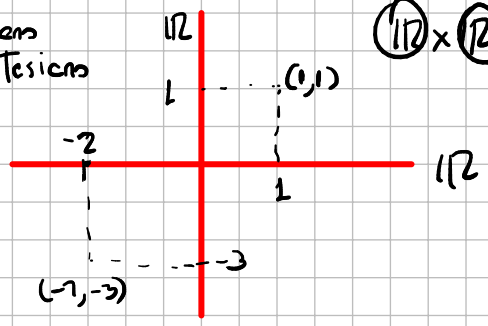
⑤ $A = \{x \in \mathbb{R} / 2 < x < 5\} \subseteq \mathbb{R}$

$B = \{y \in \mathbb{R} / -1 < y \leq 3\} \subseteq \mathbb{R}$

"$\wedge$" significa "y"; "$\vee$" significa "o".

$A \times B = \{(x,y) \in \mathbb{R} \times \mathbb{R} / \underline{2 < x < 5} \wedge -1 < y \leq 3\}$

Plano Cartesiano



$\mathbb{R} \times \mathbb{R}$

Prod. Cartesiano: $A \times B$

$A \times B = \{(x,y) / x \in A \wedge y \in B\}$

$A = \mathbb{R}$

$B = \mathbb{R}$

$A \times B = \mathbb{R}^2$ plano cartesiano

Compacto: $\underset{(-3;5]}{A} \subseteq \mathbb{R} \wedge \underset{\{4\}}{B} \subseteq \mathbb{R}$ $\quad -3 < x \leq 5$

$A \times B = \{\underline{(x,y) \in \mathbb{R} \times \mathbb{R}} / x \in (-3; 5), y = 4\}$ $\quad (-3; 5]$

$= \{(x,y) \in \mathbb{R} \times \mathbb{R} / x \in (-3; 5]\}$

Expandida:

$A = \{1, 2\}$ $\quad A \times B = \{(1;3), (1;4), \ldots\}$

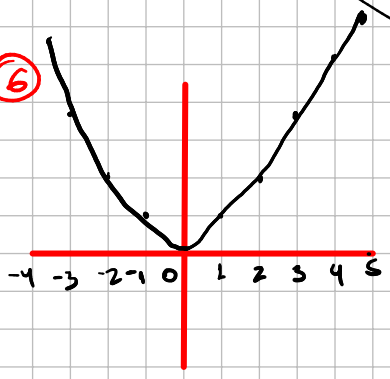
$B = \{3, 4\}$

$A = (2; 5)$ $\quad A \times B = \{(2.5, 1), (3; 1), (\pi; 1)$

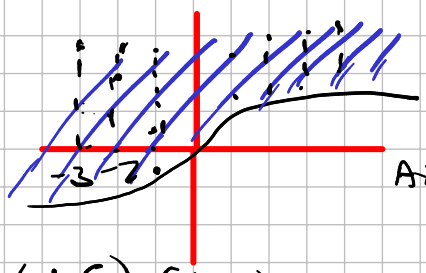
$B = \{2\}$ $\quad (e; 1), (2.9; 1), (2.99, 1)$

$(2.999, 1)\ (2.9999, 1)\}$

⑥



$R = \{\underline{(x,y) \in A \times B} / \underline{y^3 \geq x}\} \subseteq A \times B$



$A = \{-1, 0, 1, -2, -3, 3, 2, 5\}$

$B = \{-1, 14, 5, 7, 12, 13\}$

$A \times B = \{(-1;-1), (-2;-1); (-3,-1), \ldots$

$\underbrace{(\ldots; 14), (\ldots, 5), (\ldots, 7)}_{\text{función}}$

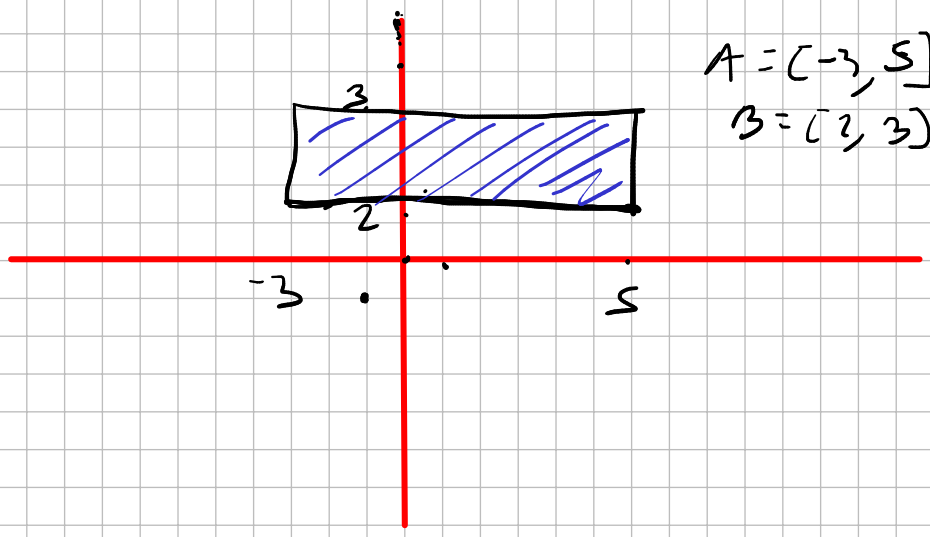
$R = \{(-1,-1), (-1,14), (-1,5), (-1,\ldots),$

$(0,14), (0,5), \ldots$

$(1,14), \ldots$

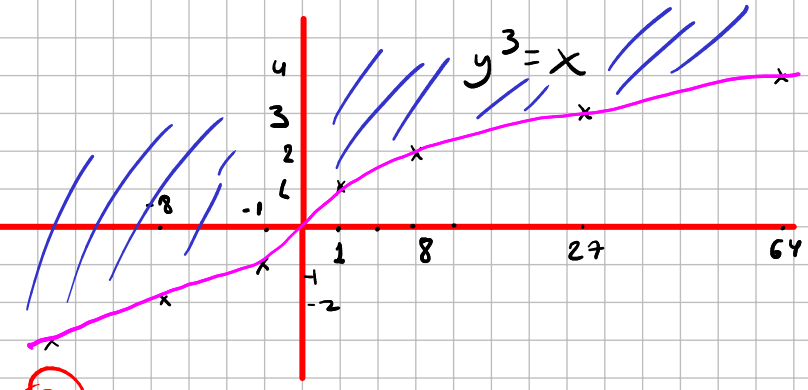
$(-2,\ldots), (3,\ldots), (2,\ldots), (5,\ldots)\}$

$(-3,\ldots)$

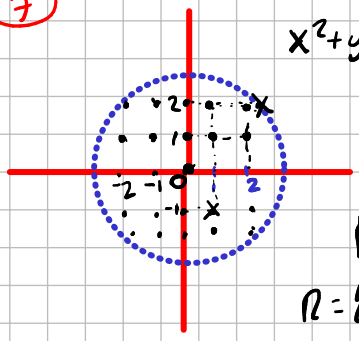


$A = [-3, 5]$

$B = [2, 3]$



(7)



$x^2+y^2 < 9 = (3)^2$

$A \times B = \{$ ////////// $\}$

$R = \{(x,y) \in A \times B \,/\, x^2+y^2<9\}$

$R = \{(-1,1), (0,-1), (1,-1), (-2,-1), (2,-1)\}$

Dominio y Rango:

R es una función

$Dom(R) = \{-1, 0, 1, -2, 2\}$

$Ran(R) = \{1, -1\}$

(10) T es relación $\mathbb{R}$: $T \subseteq \mathbb{R} \times \mathbb{R}$

$(x,y) \in T \iff k^2 - kx + x^2 = 4 + ky - y^2$

reflexividad:

Hallar todos los $x \in \mathbb{R}$ tales que $(x,x) \in T$

$k^2 - kx + x^2 = 4 + kx - x^2$

$2x^2 - 2kx + k^2 - 4 = 0$

$x^2 - kx + \frac{k^2-4}{2} = 0$

$a = 1, b = -k, c = \frac{k^2-4}{2}$

$x = \frac{-(-k) \pm \sqrt{(-k)^2 - 4(1)\left(\frac{k^2-4}{2}\right)}}{2(1)} = \frac{k \pm \sqrt{8-k^2}}{2}$

• Para que $x \in \mathbb{R}$: $\sqrt{8-k^2} \in \mathbb{R}$

$8 - k^2 \geq 0 \to 8 \geq k^2$

$\sqrt{8} \geq \sqrt{k^2} = |k|$

$-\sqrt{8} \leq k \leq \sqrt{8}$ Sol: $[-\sqrt{8}, \sqrt{8}]$

simétrica: Tomando cualquier $k \in \mathbb{R}$

Tomando $(x,y) \in T$, encontrar los $(y,x) \in T$:

$\begin{cases} k^2 - ky + y^2 = 4 + kx - x^2 \iff (y,x) \in T \\ k^2 - kx + x^2 = 4 + ky - y^2 \iff (x,y) \in T \end{cases}$

$k^2 - 4 - kx + x^2 = ky - y^2$

$\cancel{k^2} - \cancel{kx} + x^2 = \cancel{k^2} - 4 - \cancel{kx} + x^2 + 4$

$\cancel{x^2} = -4 + \cancel{x^2} + 4$

$\cancel{x^2} = \cancel{x^2}$

$0 = 0$

Hallar y:

$\cancel{k^2} - \cancel{ky} + y^2 = \cancel{4} + \cancel{k^2} - \cancel{4} - \cancel{ky} + y^2$

$\cancel{y^2} = \cancel{y^2}$

$0 = 0$

! $\Rightarrow$ Entonces se cumple: Si $(x,y) \in T$ $\Rightarrow (y,x) \in T$ (i.e.: T es simétrica y además $\forall k \in \mathbb{R}$) "es decir"

Por ejemplo:

Si $k = 0$:

$(x,y) \in T \iff x^2 = -y^2$

$x - 1 = 0$ Tenemos un $(x,y) \in T$: $x^2 = -y^2$

$(x-1)^2 = 0$ $\qquad y^2 = -x^2$

$\parallel$ $\qquad (y,x) \in T$

$x^2 - 2x + 1 = 0$

Si $(x,y) \in T \Rightarrow (y,x) \in T$

$x = 1 \Leftrightarrow$ S.I: $k \in \mathbb{R}$

$\hookrightarrow x \neq 1$:

$x^2 y - 2xy + y - x = 0$

$(x^2 - 2x + 1)y - x = 0$

$(x-1)^2 y - x = 0$

Si $x < 1$: $y = \frac{x}{(x-1)^2} = \frac{1/2}{\frac{1}{4}} = \frac{4}{2} = 2$

$y = \frac{3/4}{(1/4)^2} = \frac{3/4}{1/16} = \frac{16 \cdot 3}{4} = 12$

$x = 1/2$ $\qquad x = 3/4$ $\qquad x = -1$

$y = \frac{-1}{(-1-1)^2} = \frac{-1}{4}$

$x = -3$

$y = \frac{-3}{(-3-1)^2} = \frac{-3}{16}$

$x \leq y \iff x < y \vee x = y$

$0 = \frac{1}{y}$

$1 = y \cdot 0$

$1 = 0$

$\Leftrightarrow$

$0 < (x-1)^2 = \frac{x}{y}$

(12)

Si $x > 1$:

$$x = 2$$

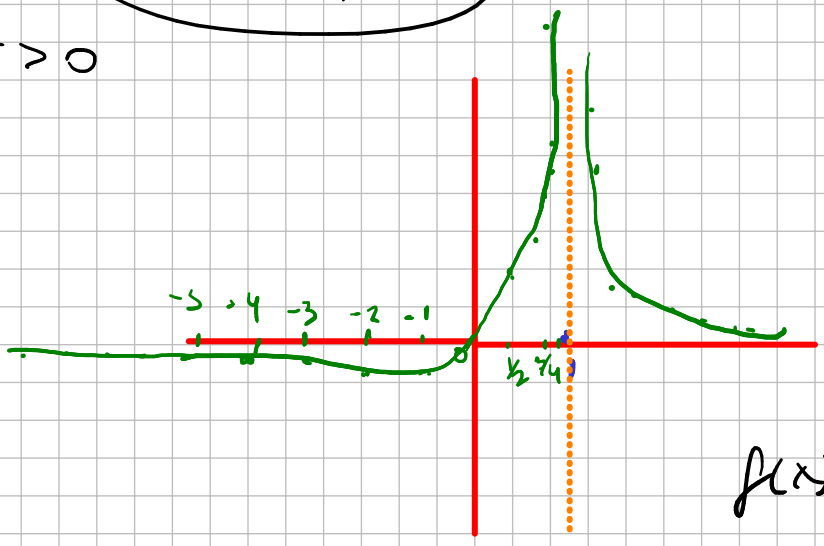
$$y = \frac{2}{(2-1)^2} = 2$$

$$(x-1)^2 \geq 0$$

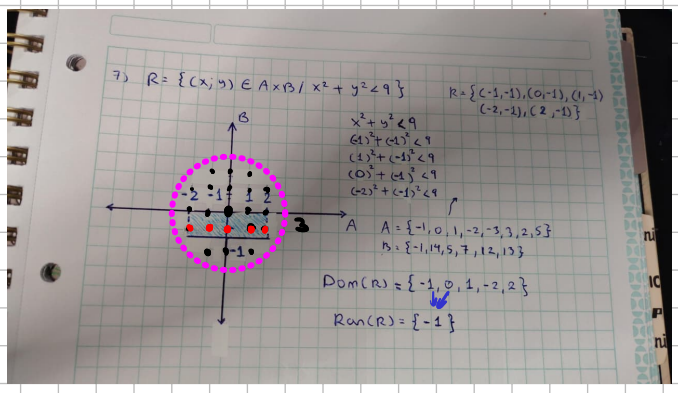
$$(x-1)^2 = 0 \iff x-1=0 \iff x=1$$

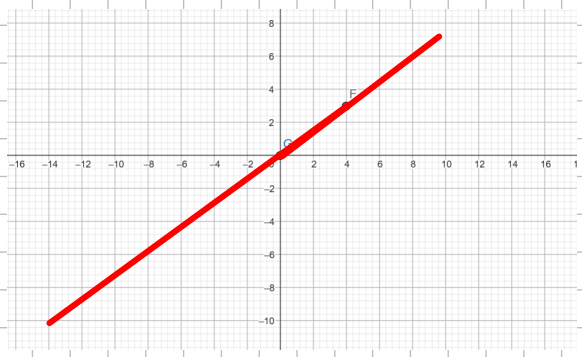
$$x \neq 1 \Rightarrow (x-1)^2 \neq 0$$

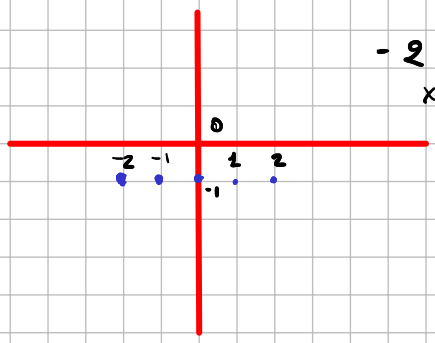
$$\Rightarrow (x-1)^2 > 0$$



$$f(x) = ax$$







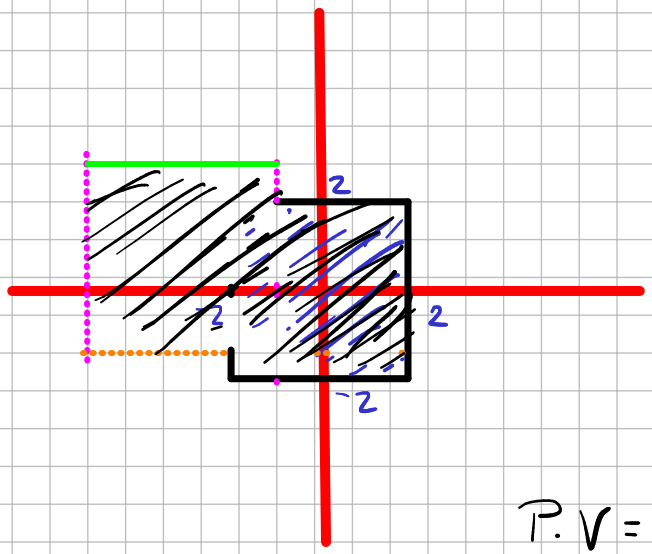
$$-2 \leq x \leq 2$$

$x = 1.5$

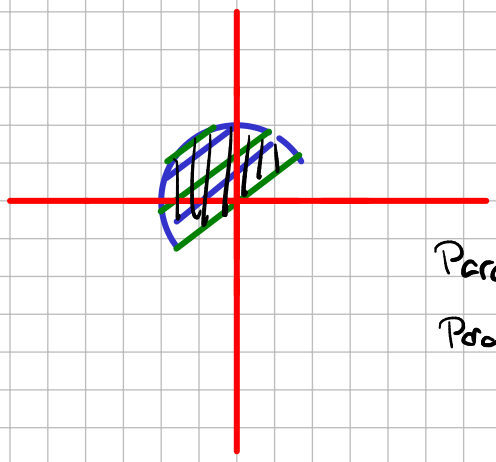
$0.5 \notin A$

-0.5

15 $(-2 \leq x \leq 2 \wedge -2 \leq y \leq 2) \vee (-5 < x < -1 \wedge -1 < y \leq 3)$



$$P.V = R\,T\,n$$



R : const.
P : Presión
V : Volúmen
T : Temperatura
n : N° moles

Es el mismo sol: Const.

$$\frac{P.V}{T} = Rn$$

Para un estado 1: P_1, V_1, T_1

Para un estado 2: P_2, V_2, T_2

$$\frac{P_1.V_1}{T_1} = Rn$$

$$\frac{P_2.V_2}{T_2} = Rn$$

$$\Rightarrow \frac{P_1.V_1}{T_1} = \frac{P_2.V_2}{T_2}$$

$P_1 : 0,75\ kg/cm^2$

$T_1 : 294\ K$

$V_1 : 8000\ cm^3$

$$\frac{P_1.V_1}{T_1} = \frac{0.75\,kg/cm^2 . 8000\,cm^3}{294\,K}$$

$T_2 = 300\,K$

$V_2 = 7000\,cm^3$

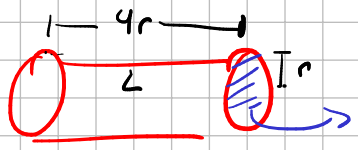
$$P_2 = \frac{P_1.V_1}{T_1} . \frac{T_2}{V_2} = \frac{0.75\,kg/cm^2 . \cancel{8000\,cm^3}^{\,4}}{\cancel{294\,K}_{\,\cancel{98}\,49}} . \frac{\cancel{300\,K}^{\,1}}{\cancel{7000\,cm^3}}$$

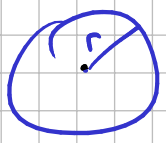
$$= \frac{75 . 4\ kg/cm^2}{49.7}$$

$$V \overset{d}{\longleftrightarrow} T \qquad \frac{V}{T} = \text{Const.}$$

$$V \overset{inv}{\longleftrightarrow} P. \qquad V.P = \text{Const.}$$

$$\frac{V.P}{T} = \text{Const.}$$





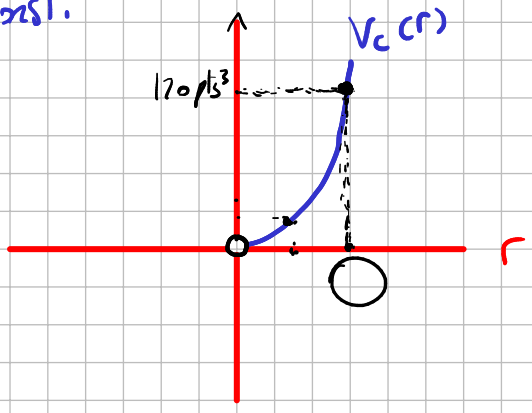
$$V_c = (\pi r^2).(4r) = 4\pi r^3$$

$r = 1$

$V_c(r) = 4\pi$

$$V_c(r) = 4\pi r^3$$

$$Dom(V_c) = \mathbb{R}^+ = \langle 0, +\infty \rangle$$



$$r^3 = \frac{V_c(r)}{4\pi} \Rightarrow r = \sqrt[3]{\frac{V_c(r)}{4\pi}}$$

Si $V_c(r) = 120\ pts^3 \Rightarrow r = \frac{\sqrt[3]{120}}{\sqrt[3]{4\pi}}\ pts = \frac{\sqrt[3]{15 \cdot 8}}{\sqrt[3]{4\pi}}\ pts = \frac{2\sqrt[3]{15}}{\sqrt[3]{4\pi}}\ pts \Rightarrow L = \frac{4(2)\sqrt[3]{15}}{\sqrt[3]{4\pi}}\ pts$

$$L = \frac{8 \cdot \sqrt[3]{15}}{\sqrt[3]{4\pi}}\ pts$$